

Quotient Laws of Exponents*Independent Set 3.02* **ANSWER KEY***Six questions, on your own, in silence. Show the step that does the work.*

1. Write as one power. $\frac{6^7}{6^3}$

$$6^4$$

2. Find the value. $\frac{5^6}{5^4}$

$$5^2 = 25$$

3. Find the value. $\frac{(-4)^5}{(-4)^3}$

$$(-4)^2 = 16$$

An even power of a negative base is positive.

4. Rewrite so the exponent is on the numerator and on the denominator. $\left(\frac{3}{7}\right)^4$

$$\frac{3^4}{7^4}$$

5. Find the value. $\left(\frac{12}{13}\right)^0$

$$1$$

Any nonzero base to the zero power is 1.

6. Find the value of n . $\frac{2^9}{2^n} = 2^4$

$$n = 5$$

$$9 - n = 4.$$